

Code-Layout, Readability and Reusability

Date	3 July 2026
Team ID	SWTID-2026-7147
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the code base in terms of its structure, readability, and reusability.

Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform indentation throughout the code.	Yes	Consistent formatting is followed.
2	Proper File Structure	Files are logically organized into folders.	Yes	Logical project structure.
3	Meaningful Variable Names	Variables use clear and descriptive names.	Yes	Easy to understand.
4	Function / Method Names	Functions are named based on their purpose.	Yes	Descriptive function names.
5	Code Comments	Comments explain important code sections.	Partial	Basic comments are provided.
6	Modular Design	Code is divided into reusable modules.	Yes	Highly modular code.
7	No Redundant Code	Duplicate and unused code is removed.	Yes	No unnecessary repetition.
8	Error Handling	Errors and exceptions are handled properly.	Yes	Errors handled appropriately.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing	Python	Model Training & Prediction	High
2	Machine Learning Models Machine Learning Models	Scikit-learn, XGBoost	Loan Prediction	High
3	Flask Web Application	Flask	User Interface	High
4	Prediction Module	Python	Real-time Loan Approval	High
5	Input Validation Module	Python	Applicant Data Processing	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured and organized project.
Readability	4	Clear naming conventions and easy to understand
Reusability	5	Modules can be reused in similar ML projects
Documentation / Comments	4	Basic comments provided; more documentation can be added
Overall Score	4.5 / 5	Well-organized, maintainable, and reusable codebase